

AYAN ALEEM

Artificial Intelligence Student

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PROFILE SUMMARY

Artificial Intelligence student with hands-on experience in Machine Learning, Computer Vision, NLP, and RAG-based AI applications. Skilled in Python, FastAPI, PostgreSQL, and LLM integration, with experience developing AI chatbots and document retrieval systems. Interested in building practical AI solutions that solve real-world problems.

EDUCATION

BS Artificial Intelligence, COMSATS University Islamabad, Attock Campus 2027 (Expected)
HSSC Pre-Engineering, Army Public College, Attock (FBISE) 2022

EXPERIENCE

IT Intern, Carrefour, Giga Mall Islamabad

- Worked on retail store systems, downloading and installing official software on staff PCs.
- Resolved technical issues on-site within enterprise IT protocols.

IT Intern, Fincon Services

- Built a chatbot using React and the OpenAI API, designing the full UI from scratch.
- Integrated OpenAI for real-time responses, adding loading states, error handling, and conversation history.
- Deployed the chatbot online with complete documentation.

PROJECTS

Smart Knowledge Chatbot with RAG *Python, FastAPI, FAISS, PostgreSQL (pgvector), Ollama, Groq API*

- Built a retrieval-augmented generation (RAG) chatbot that answers questions from uploaded PDFs and images using semantic search.
- Designed a FastAPI backend integrating PyMuPDF for PDF parsing, CLIP for image understanding, and SentenceTransformers embeddings with a FAISS/pgvector vector store.
- Implemented streaming responses, multi-turn conversation memory, multi-document support, and a custom-trained intent classifier (SentenceTransformer embeddings + Logistic Regression/SVM) to auto-route queries between document and image context.
- Migrated inference from local Ollama to the Groq API and storage from in-memory FAISS to PostgreSQL with pgvector, improving response speed and persisting data across restarts.

Arabic Text Classification System *Python, Scikit-learn, PyTorch, Streamlit*

- Trained 6 models across 3 categories — 2 machine learning (Logistic Regression, SVM), 2 deep learning (LSTM, CNN), and 2 transformer-based LLMs (AraBERT, mBERT) — for Arabic text classification.
- Compiled a labeled training dataset from the ARPD and AJGT sources.
- Deployed a Streamlit interface for real-time classification demonstrations.
- Identified and corrected a data leakage issue caused by a diacritics-based feature, improving evaluation reliability.

House Price Prediction *Python, Scikit-learn*

- Compared Linear Regression and Random Forest models to predict housing prices from real estate data.
- Applied feature engineering and outlier removal during data preprocessing.
- Measured model accuracy using RMSE, MAE, and R².

CERTIFICATIONS

- Frontend Developer (React), HackerRank, 2026
- App Store Listing Certificate, Google Play Academy, 2026

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, TypeScript, SQL

AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, LLM Integration, RAG Systems, TensorFlow, PyTorch, Scikit-learn, OpenCV, MediaPipe, Hugging Face

Backend & Data: FastAPI, PostgreSQL, pgvector, FAISS, REST APIs, Streamlit

LLM & AI Tools: Groq API, Google Gemini API, Ollama, Sentence Transformers, Embeddings, Prompt Engineering

Frontend: React, Tailwind CSS, Framer Motion, Vite, HTML, CSS

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Anaconda

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Networks

Languages Spoken: English (Fluent), Urdu (Fluent)